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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Student Roll No: 14

Experiment – 3 : Infrastructure as A Service

Aim: To study and Implement Infrastructure As a Service using different cloud platform AWS / GC / AZURE (Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to –

- Launch an instance
- install any security updates
- execute bootstrap script
- EC2 instance store life
- access the instance metadata from the OS
- Amazon EBS Volume life

Lab objective mapped : ITL802 : To implement IAAS services.

Prerequisite: Concept of Operating System, Infrastructure.

Requirements: Cloud Login, Desktop, Browser, Internet etc.

Pre-Experiment Theory:

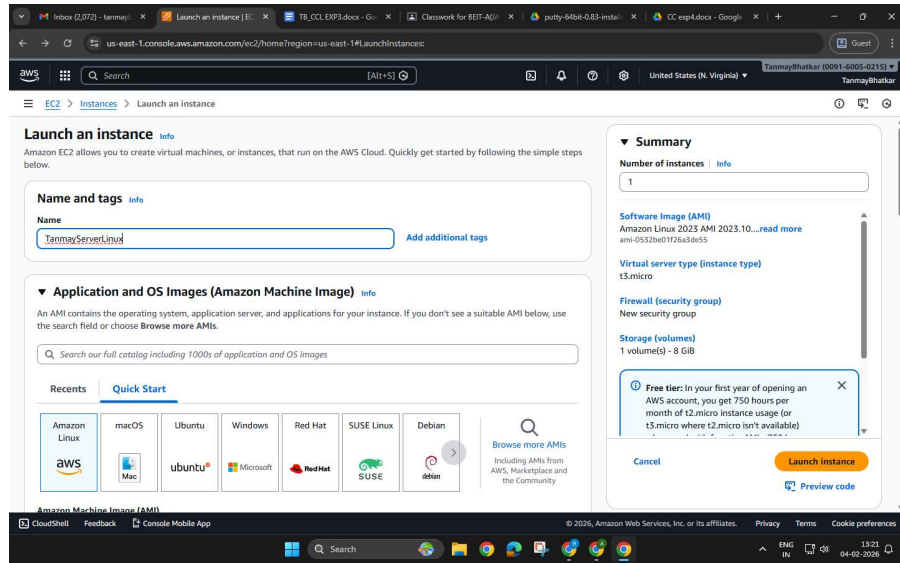
CLOUD SERVICES

IAAS.

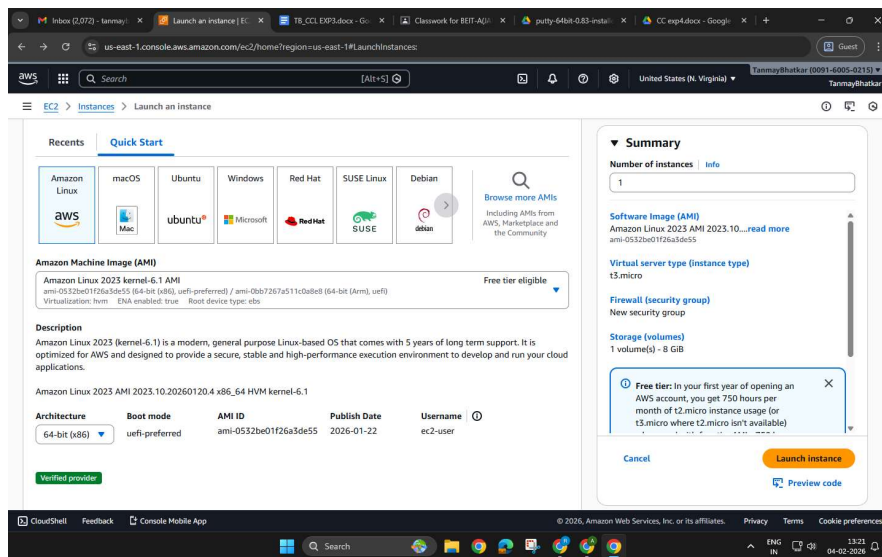
Procedure

Launch and Connect to a Linux Instance, log in with SSH, and install any security updates.

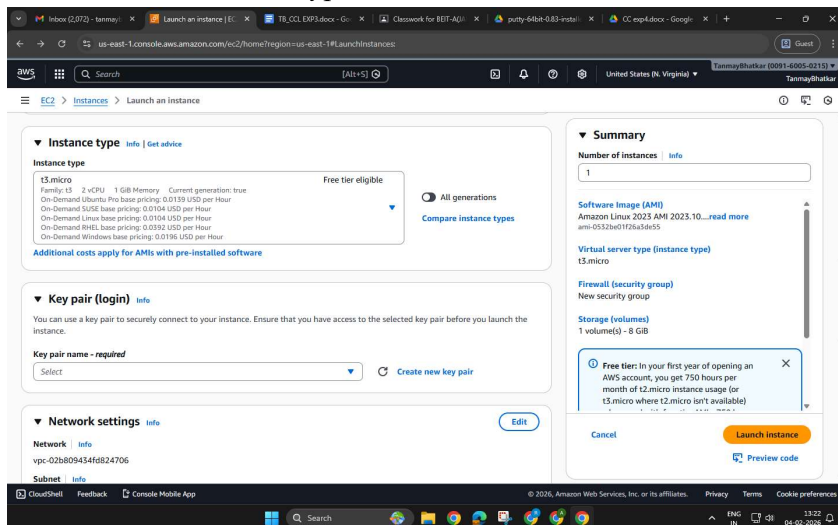
1. Launch an instance in the Amazon EC2 console.

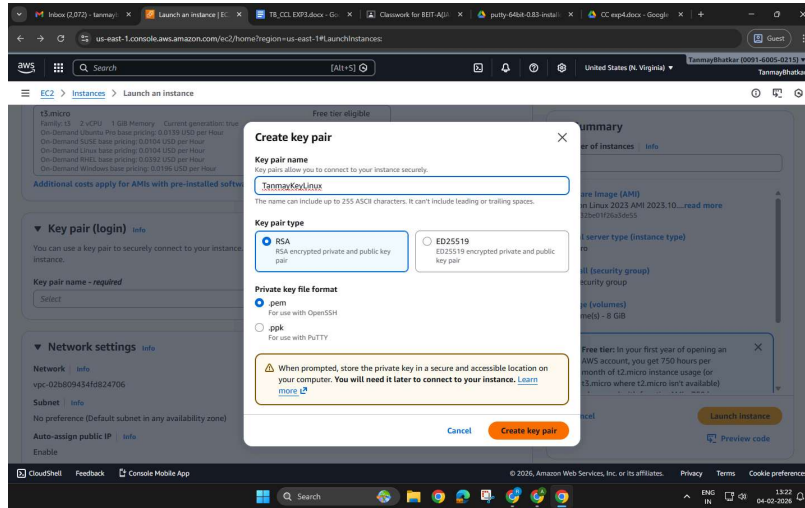


2. Choose the Amazon Linux AMI.

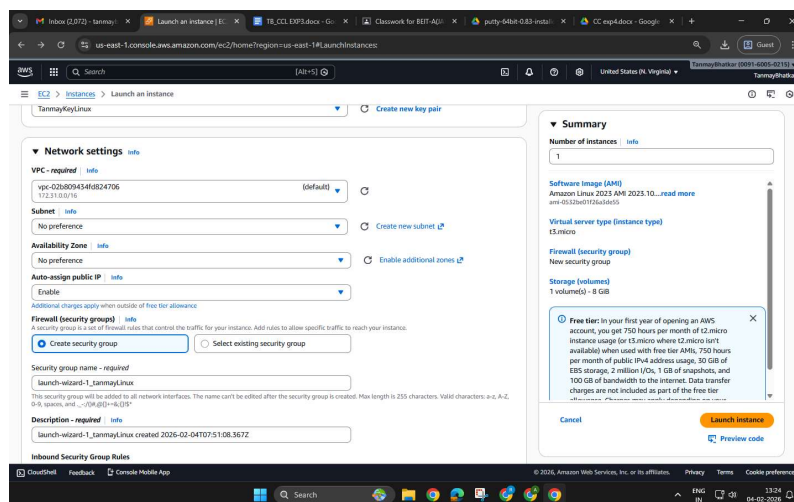
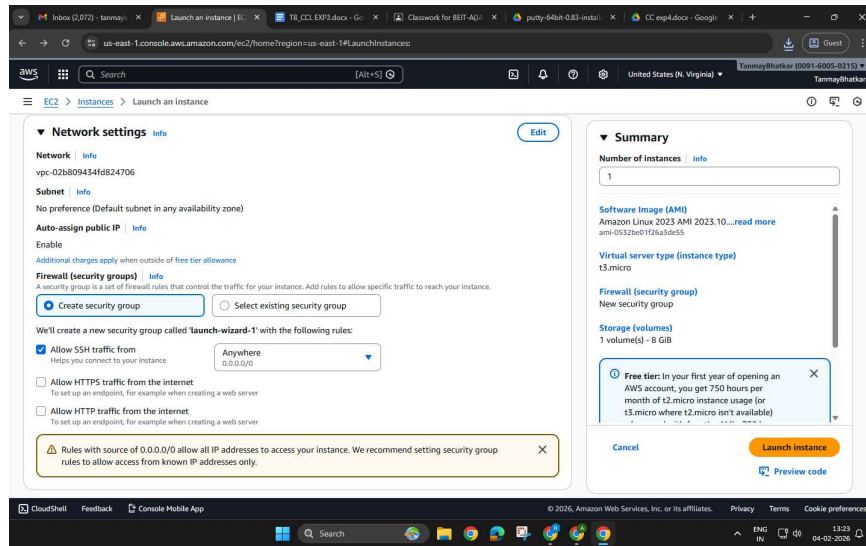


3. Choose the t2.micro instance type.

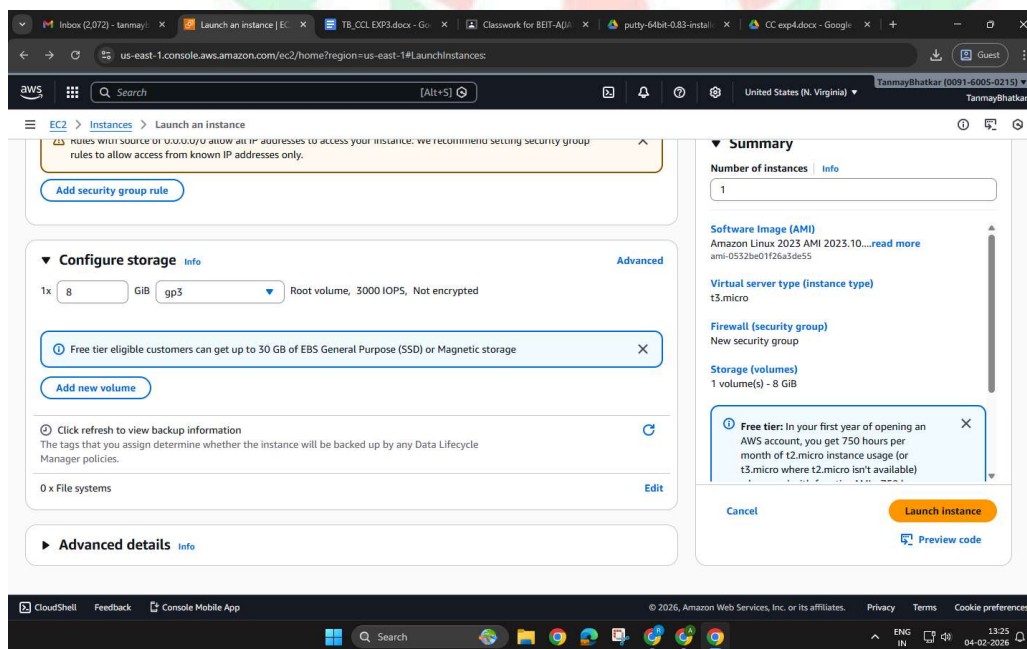
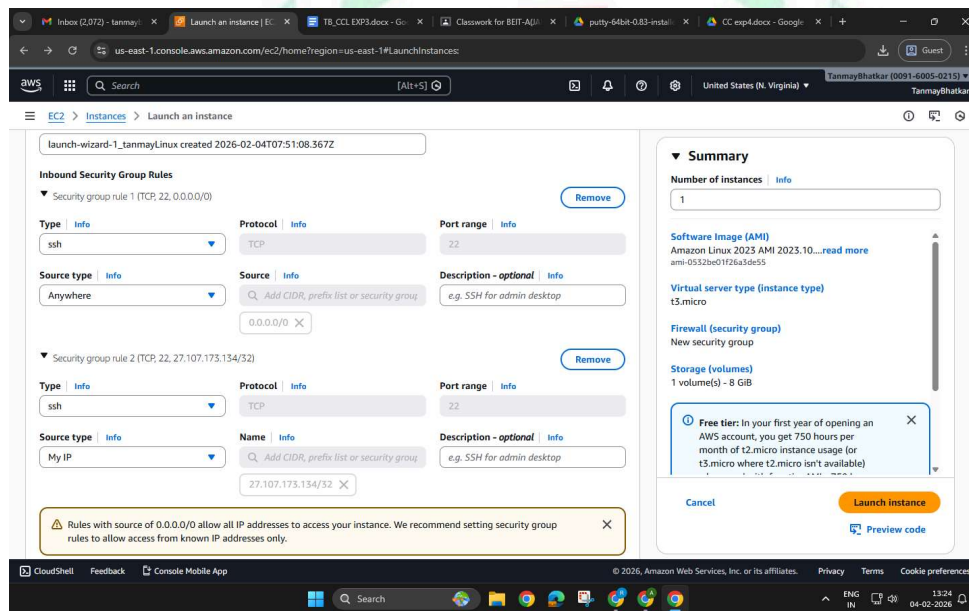




4. Launch the instance in the default VPC.



5. Assign the instance a public IP address.

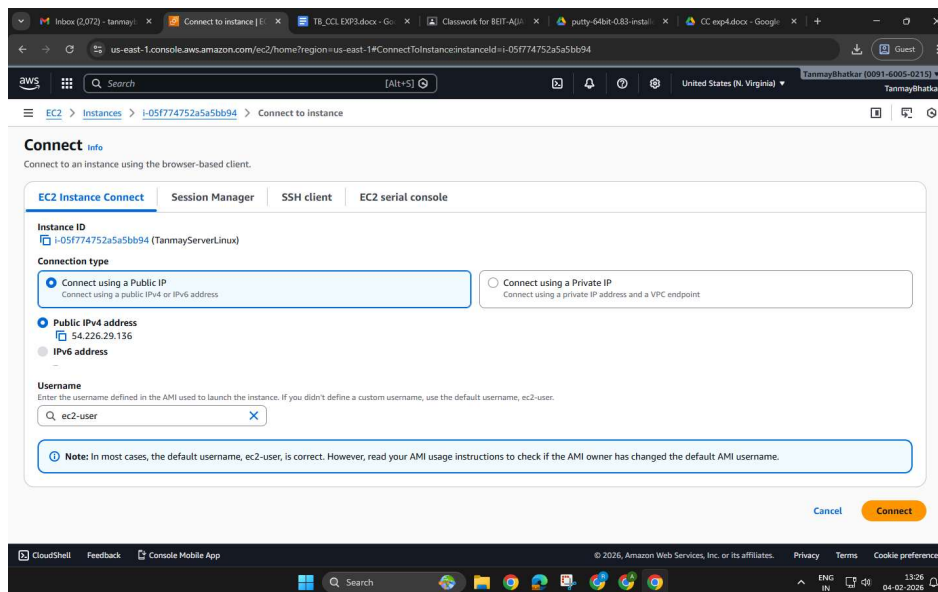
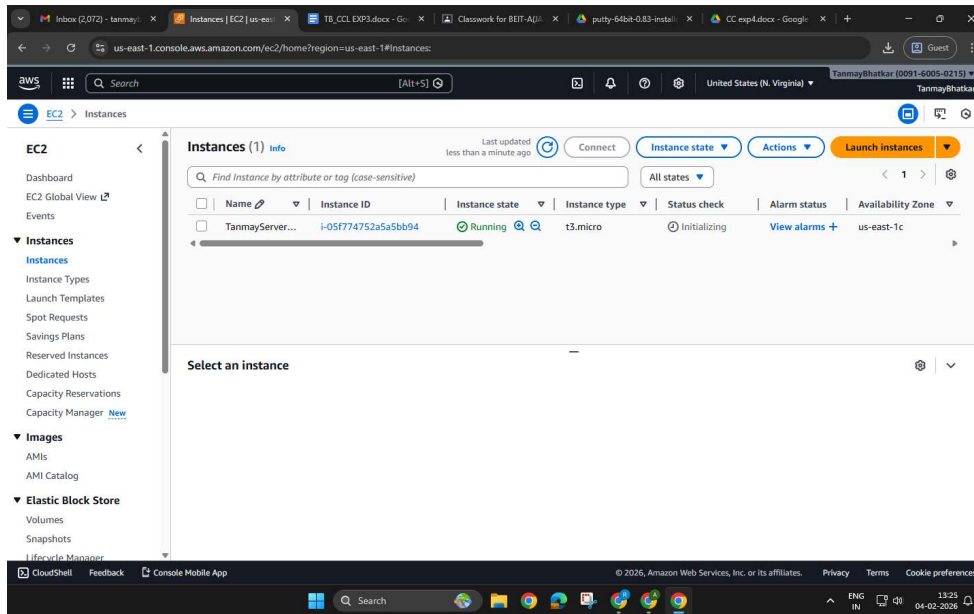


6. Add a tag to the instance of Key: Name, Value: Exercise 3.1.

7. Create a new security group called Cert Book.

8. Add a rule to Cert Book allowing SSH access from the IP address of your workstation (www.WhatsMyIP.org is a good way to determine your IP address).

9. Launch the instance.



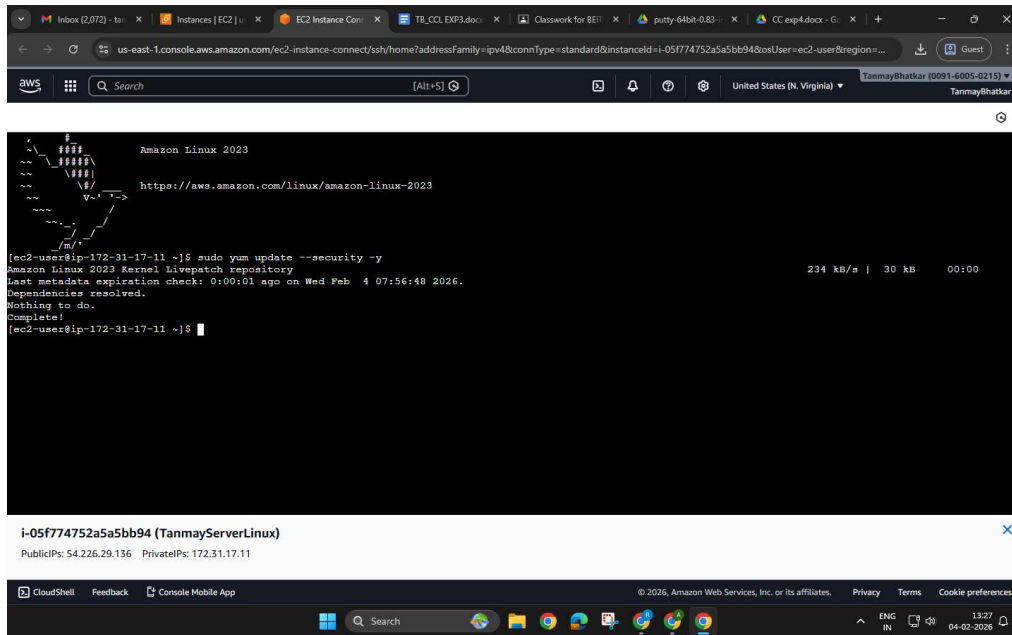
Connecting to instance using EC2 instance connect

10. When prompted for a key pair, choose a key pair you already have or create a new one and download the private portion.

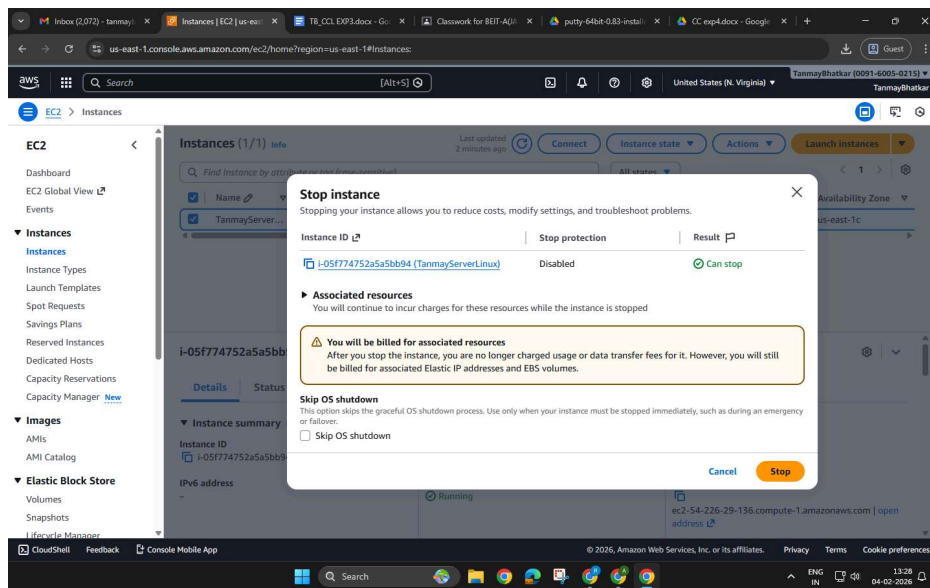
Amazon generates a keyname.pem file, and you will need a keyname.ppk file to connect to the instance via SSH. Puttygen.exe is one utility that will create a .ppk file from a .pem file.

11. SSH into the instance using the public IP address, the user name ec2-user, and the keyname.ppk file.

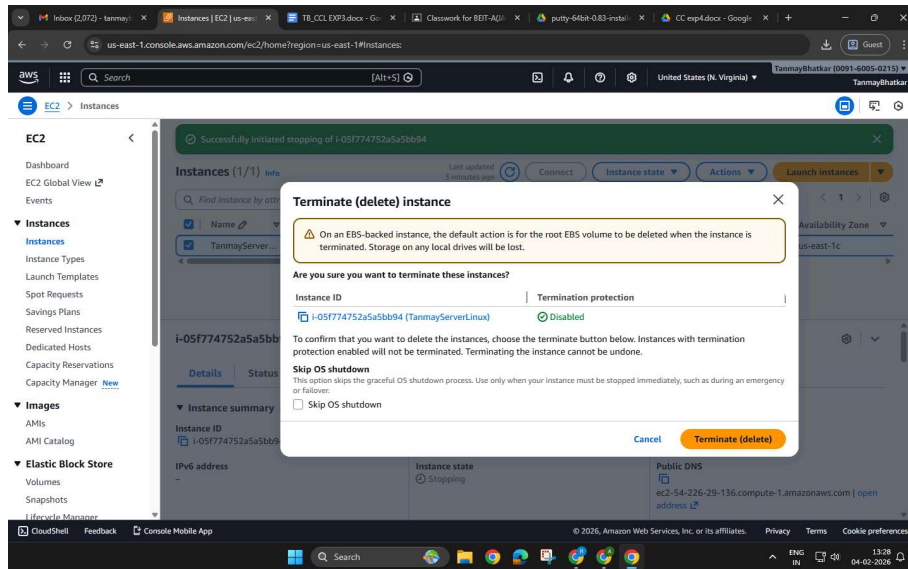
12. From the command-line prompt, run `sudo yum update—security -y`.



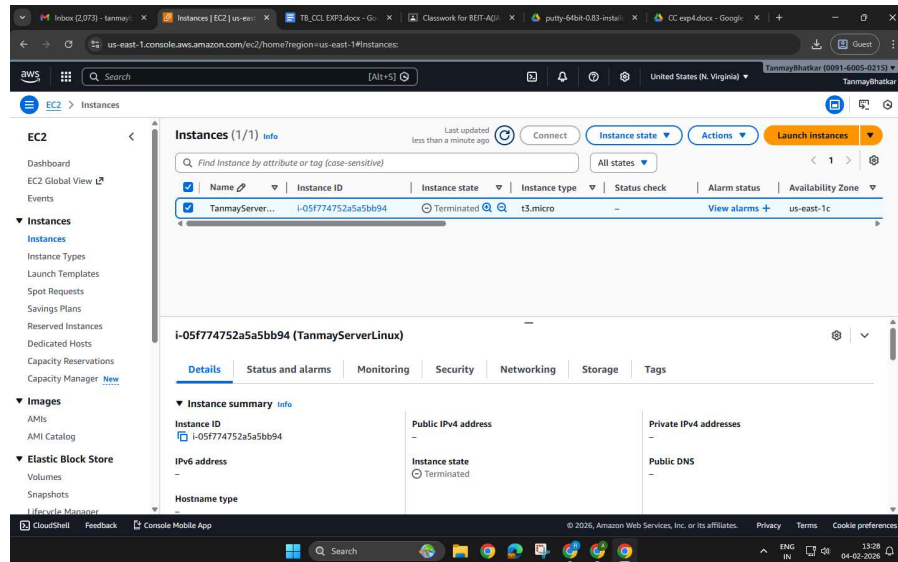
13. Close the SSH window.



Stopping instance



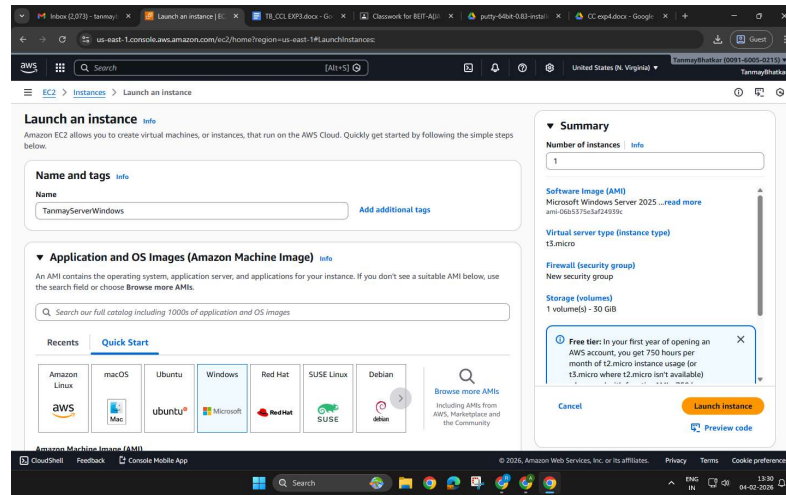
Terminating instance



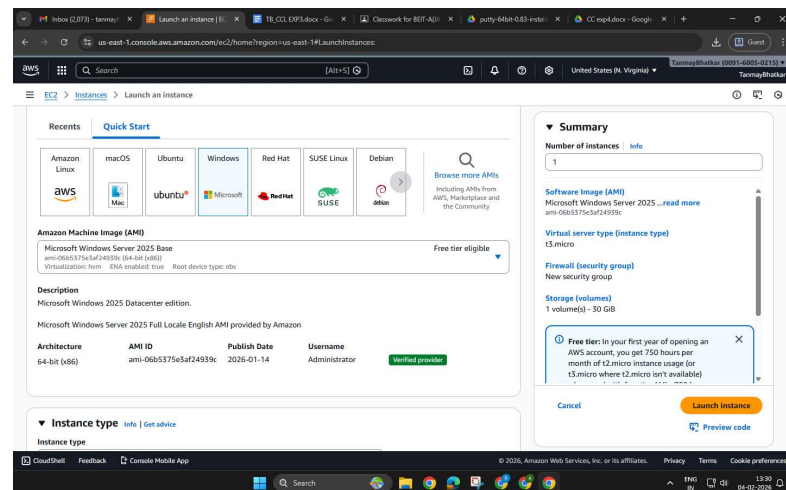
Launch a Windows Instance with Bootstrapping

Specify a very simple bootstrap script. then confirm that the bootstrap script was executed on the instance.

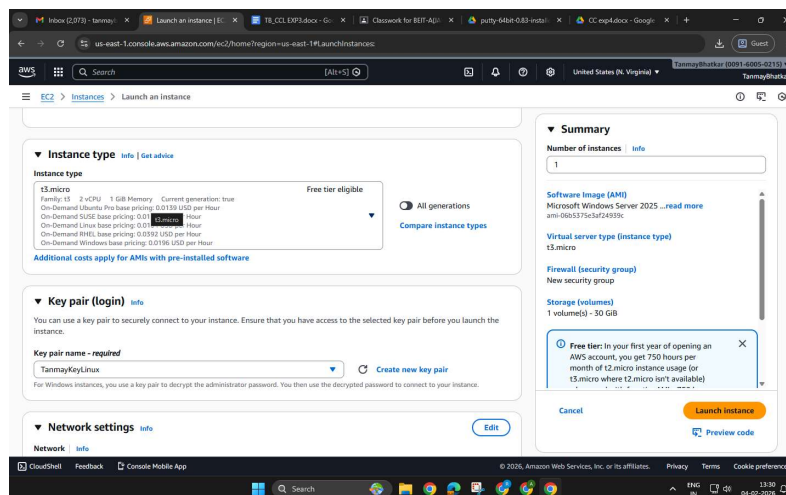
1. Launch an instance in the Amazon EC2 console.



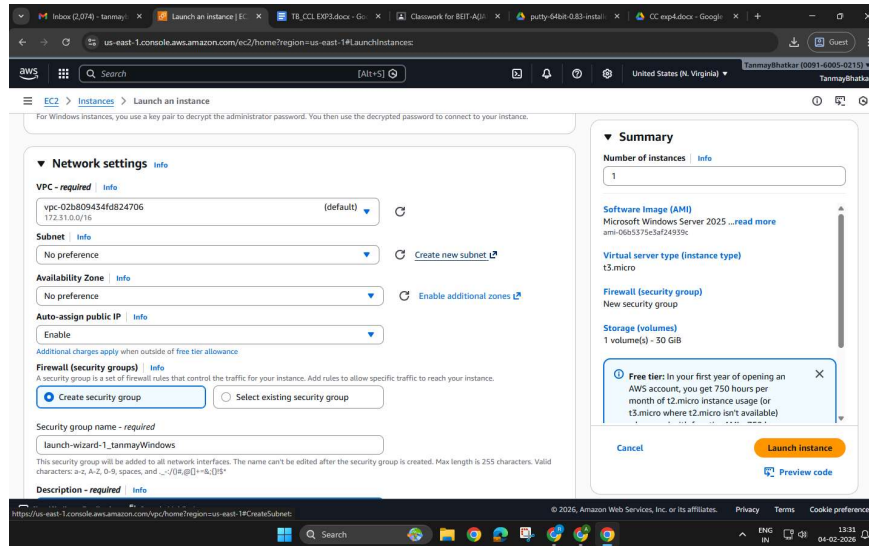
2. Choose the Microsoft Windows Server 2012 Base AMI.



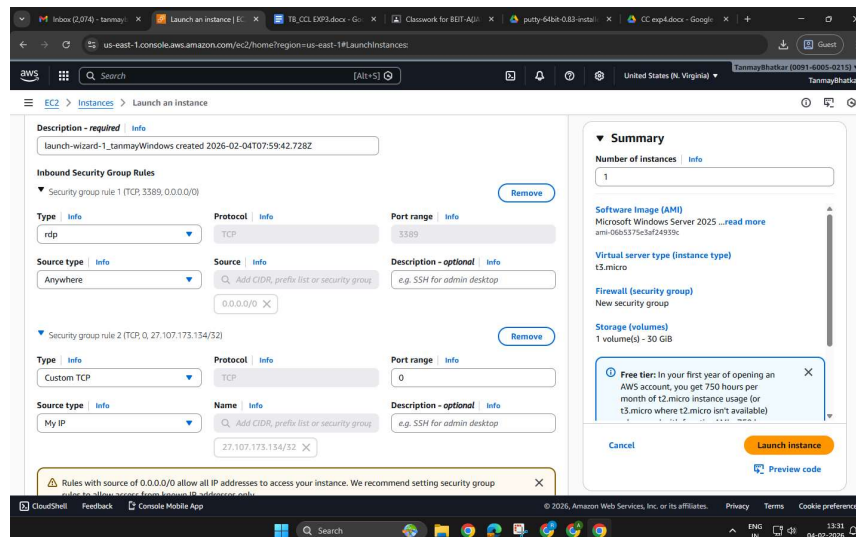
3. Choose the t2.micro instance type.



4. Launch the instance in either the default VPC.



5. Assign the instance a public IP address.

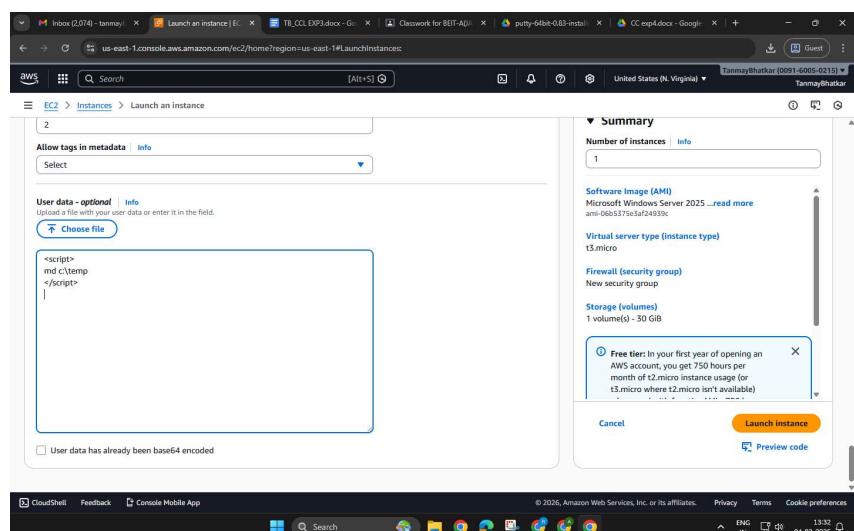


6. In the Advanced Details section, enter the following text as UserData:

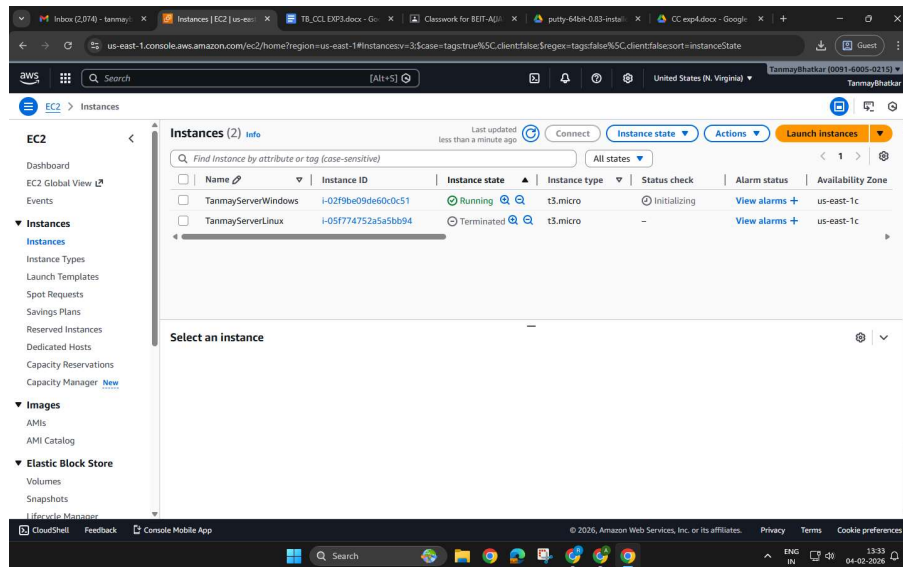
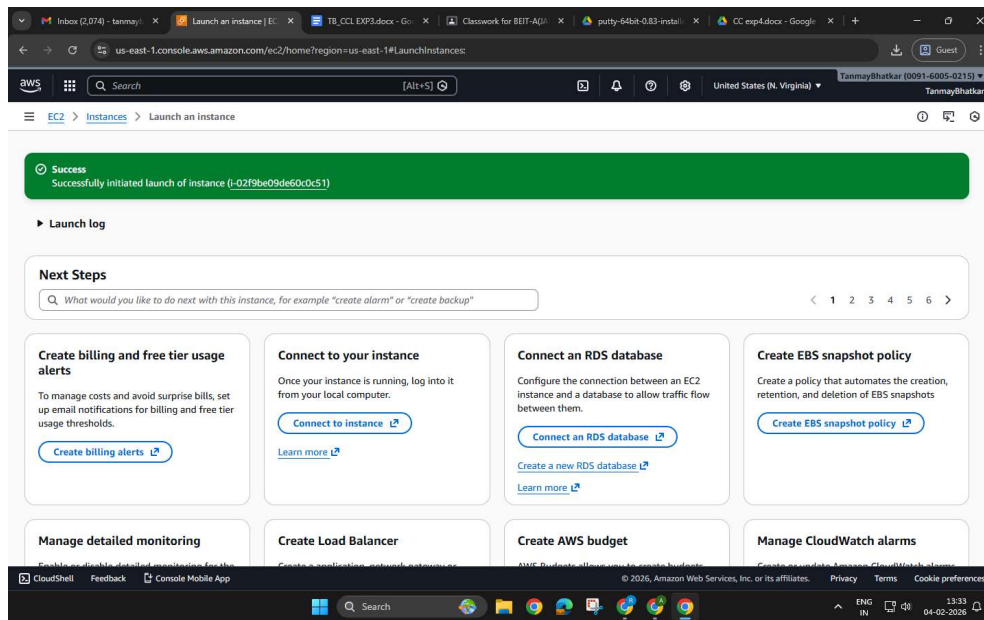
```
<script>
md c:\temp
</script>
```

7. Add a tag to the instance of Key: Name, Value: Exercise 3.2.

8. Use the Cert Book security group from Exercise 3.1.



9. Launch the instance.

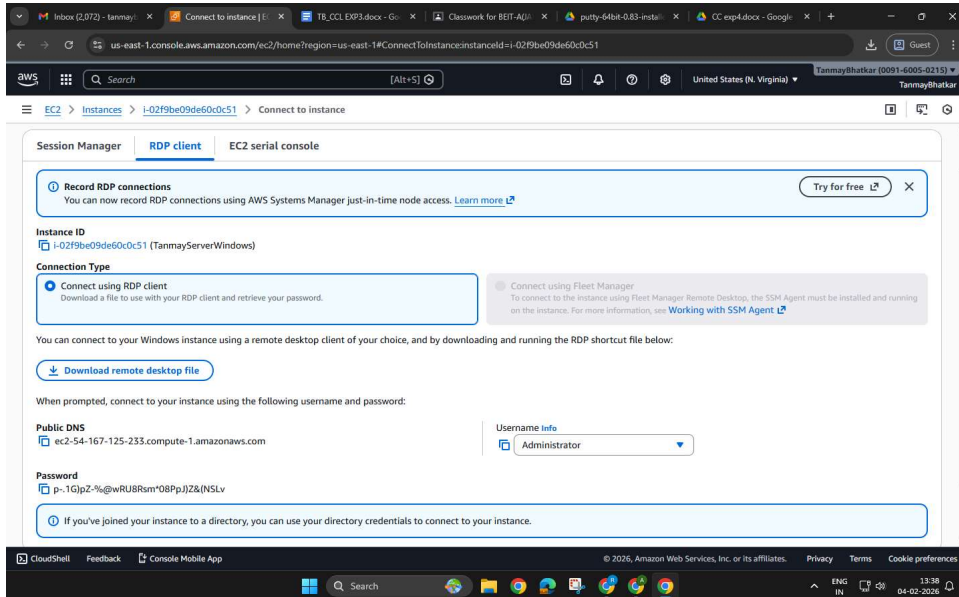
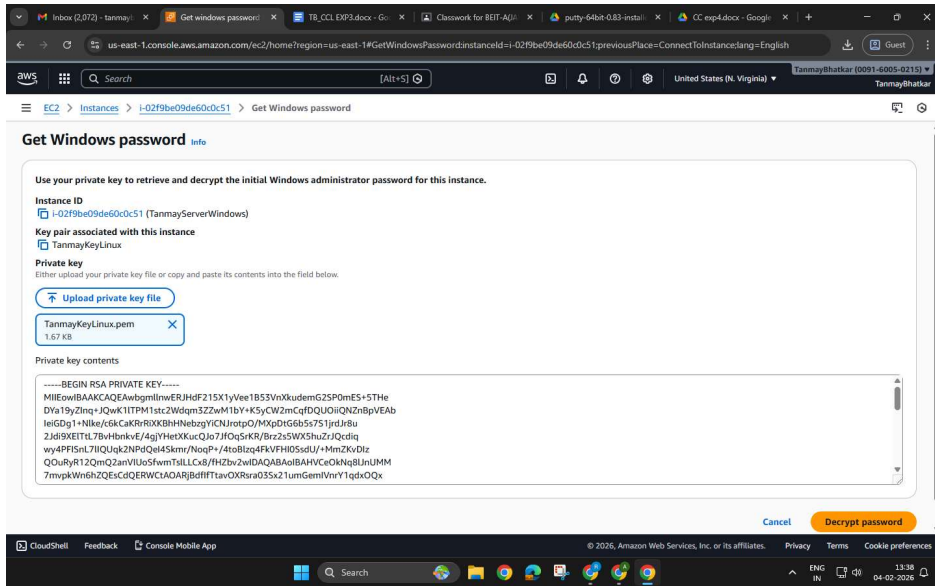
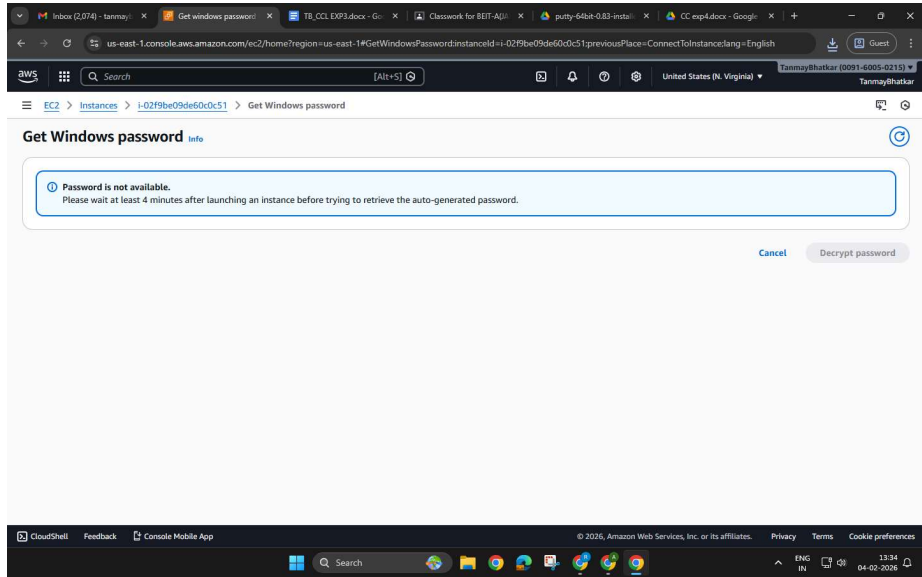


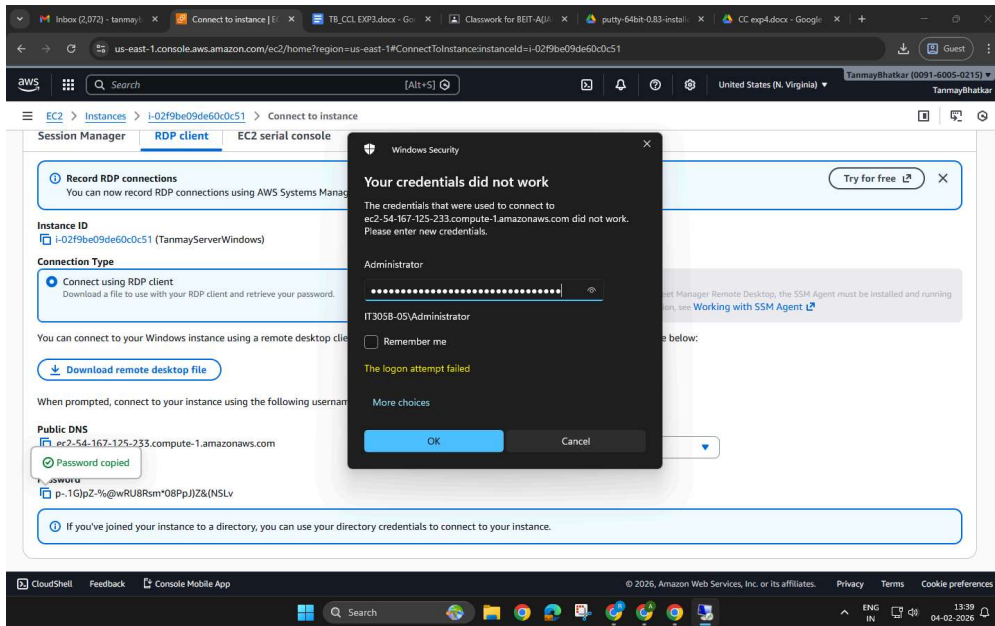
10. Use the key pair from Exercise 3.1.

11. On the Connect Instance UI, decrypt the administrator password and then download the RDP file to attempt to connect to the instance. Your attempt should fail because the Cert Book security group does not allow RDP access.

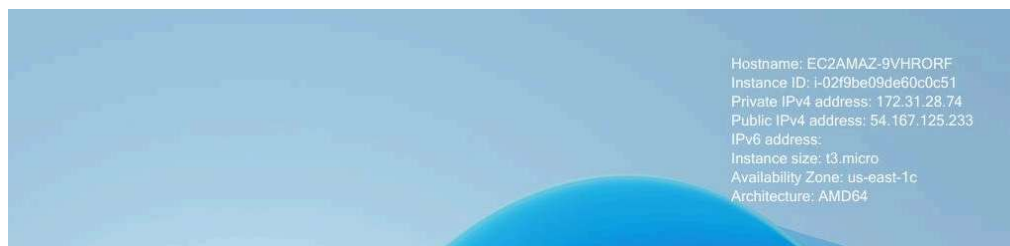
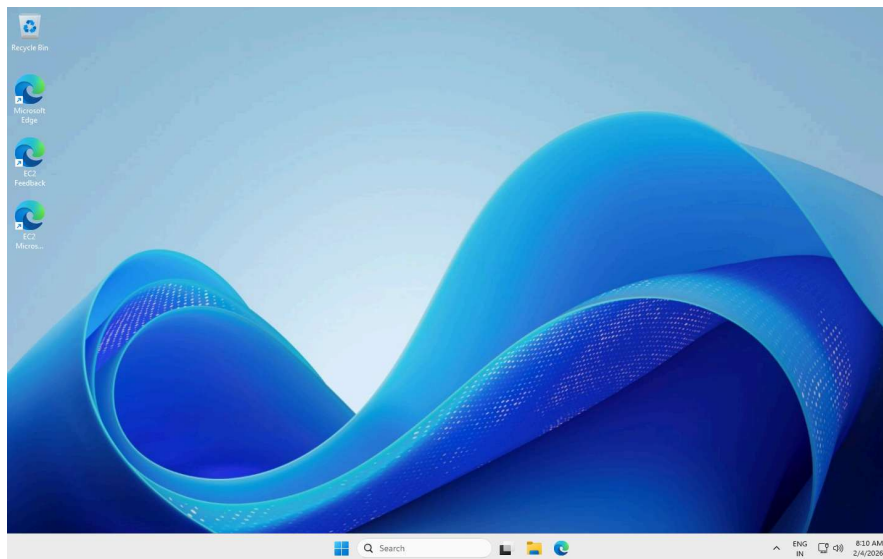
12. Open the Cert Book security group and add a rule that allows RDP access from your IP address.

13. Attempt to access the instance via RDP again.





Connecting using RPC

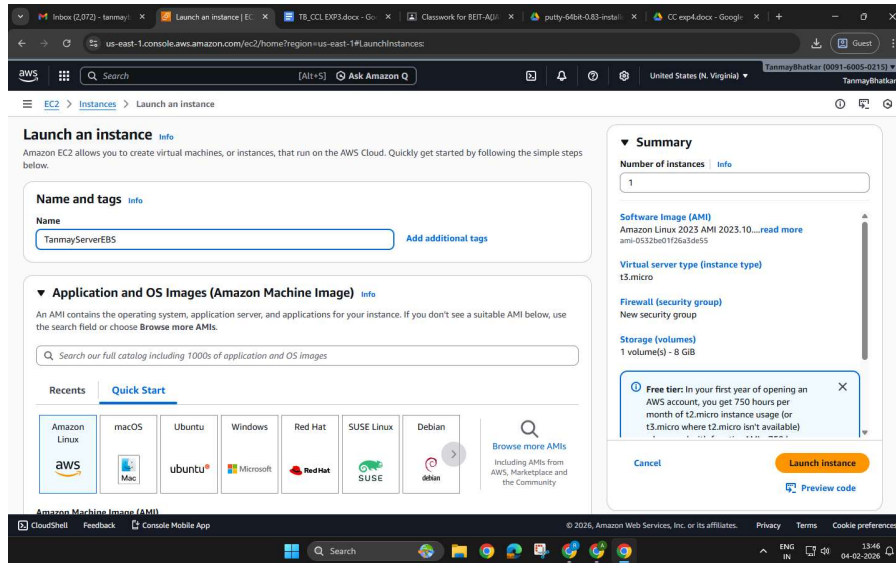


14. Once the RDP session is connected, open Windows Explorer and confirm that the c:\temp folder has been created.
15. End the RDP session and terminate the instance.

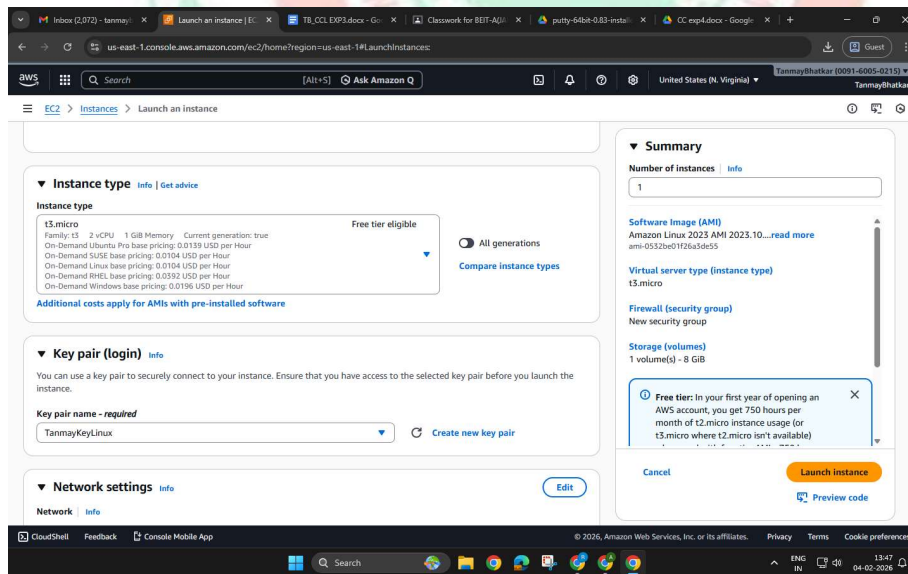
Post-Experiments Exercise

Create an Amazon EBS Volume and observe that it remains after the instance is terminated. Amazon EBS volume persists beyond the life of an instance.

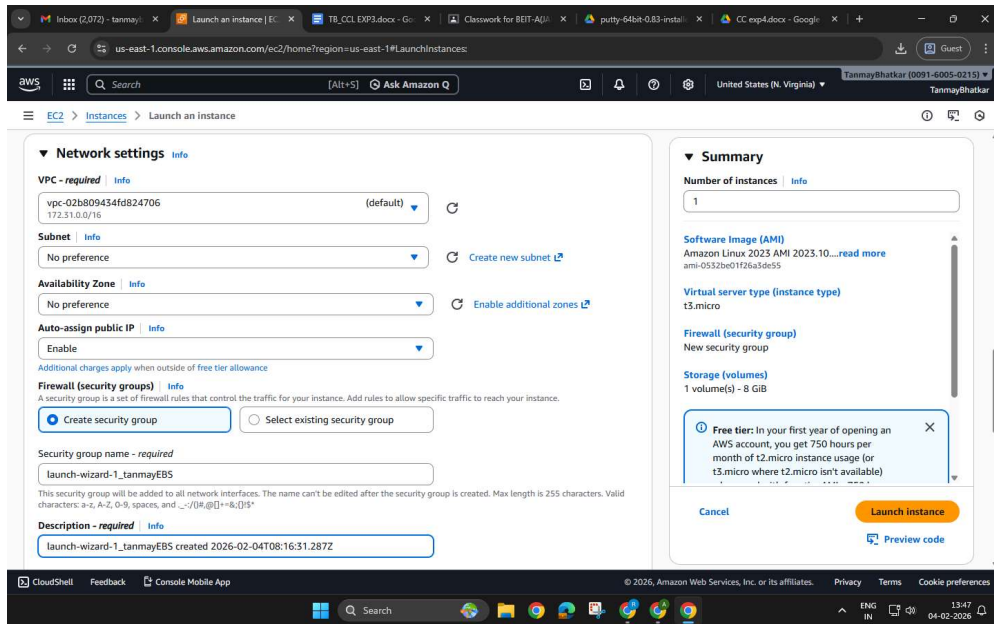
1. Launch an instance in the Amazon EC2 console.
2. Choose the Amazon Linux AMI.



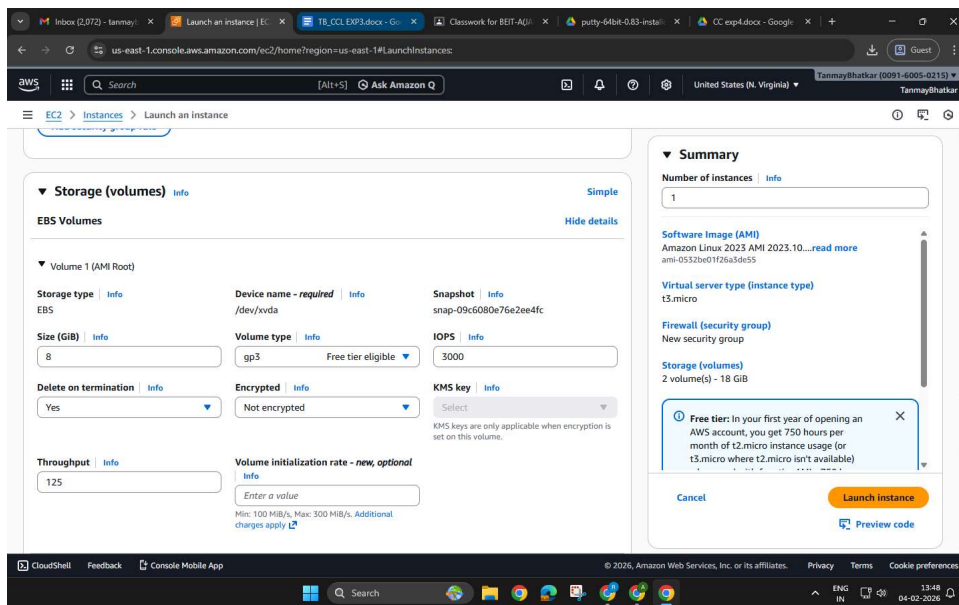
3. Choose the t2.micro instance type.

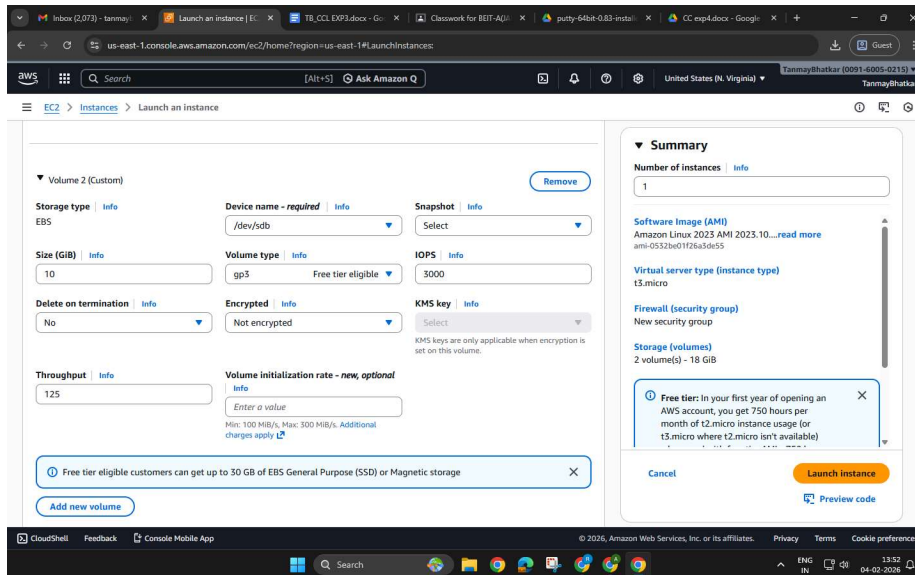


4. Launch the instance in the default VPC.



5. Add a second Amazon EBS volume of size 10 GB. Note that the Root Volume is set to Delete on Termination.





7. Add a tag to the instance of Key: Name, Value

8. Use the Cert Book security group from earlier exercises.

9. Launch the instance.

10. Find the two Amazon EBS volumes on the Amazon EBS console. Name them both

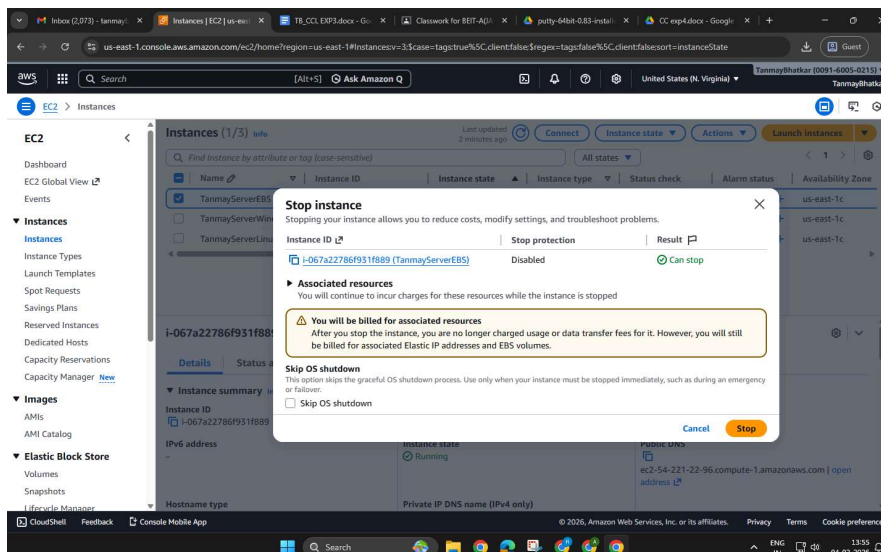
Exercise 3.6.

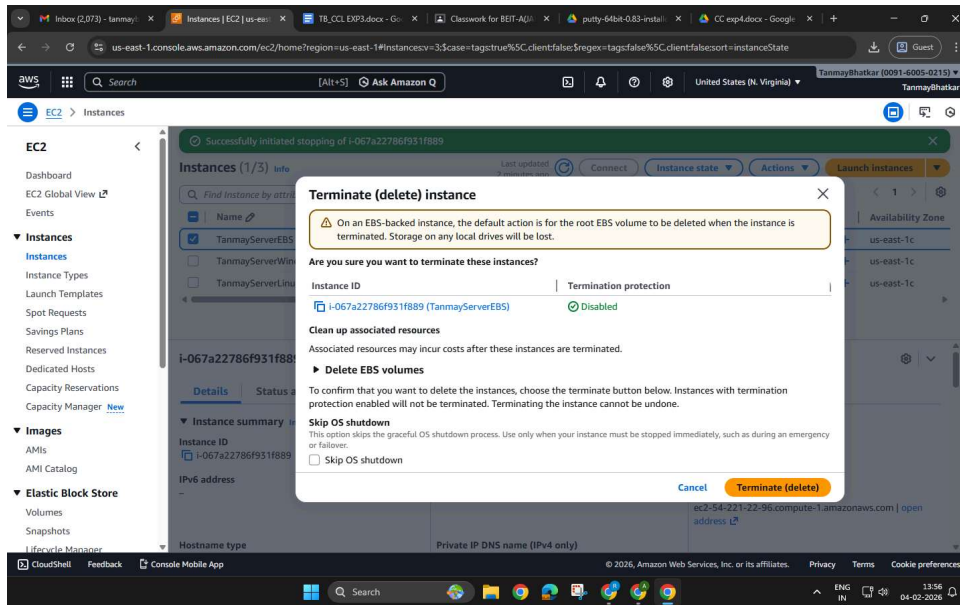
Volumes (2) [Info](#) Last updated 1 minute ago [Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets [Choose filter set](#)

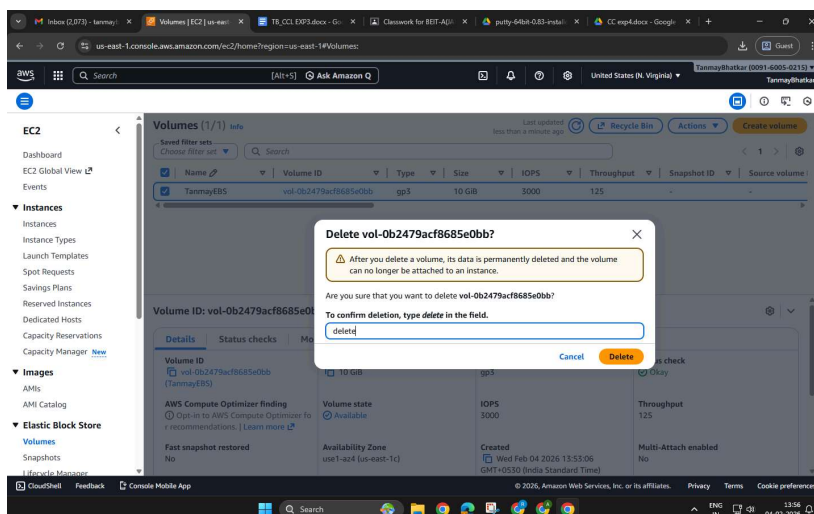
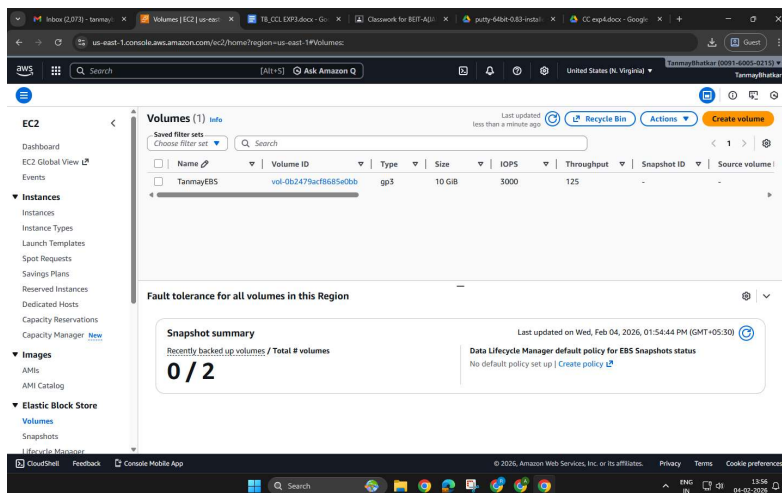
☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume
☐	TanmayEBS	vol-0b2479acf8685e0bb	gp3	10 GiB	3000	125	-	-
☐	TanmayRoot	vol-0730bc78b46636dbf	gp3	8 GiB	3000	125	snap-09c6080...	-

11. Terminate the instance.





Notice that the boot drive is destroyed, but the additional Amazon EBS volume remains and now says Available. Do not delete the Available volume.



Extended Theory:

1. What is bootstrap in a computer? (to be written in hand)
2. RDP (to be written in hand)
3. Discuss procedure to connect to your Linux instance using PuTTY (soft copy form)

Procedure to Connect to Linux Instance Using PuTTY

1. **Install Required Tools**
 - Install **PuTTY** and **PuTTYgen** on the system
2. **Prepare the Private Key**
 - Convert the .pem key to .ppk using **PuTTYgen**.
3. **Open PuTTY**
 - Launch PuTTY and select **SSH** as the connection type.
4. **Enter Host Details**
 - Enter username@public-IP of the Linux instance.
 - Set port to **22**.
5. **Attach Private Key**
 - Go to **Connection** → **SSH** → **Auth.**
 - Browse and select the .ppk private key file.
6. **Save Session (Optional)**
 - Save the session for future use.
7. **Connect**
 - Click **Open** and accept the security alert.
8. **Access Linux Instance**
 - Successful authentication opens the Linux command-line interface.

Results/Calculations/Observations:

Fill the following observation tables-(to be written in hand)

Sr. No		
1.	SSH port no	
2.	Linux VM Public IP	
3.	t2.micro RAM and CPU capacity	
4.	Decrypted pwd length	
5.	name of security grp for windows instance	

Post Experimental Exercise-

Questions:

1. Short note on EBS volume?(soft copy form)

Amazon Elastic Block Store (EBS) is a block-level storage service provided by AWS for use with EC2 instances. It acts like a virtual hard disk that can be attached to a running or stopped EC2 instance to store data such as operating systems, applications, and databases. EBS volumes provide **persistent storage**, which means the data stored on them remains available even after the EC2 instance is stopped or restarted. They are designed for **high durability and availability** within a single Availability Zone.

EBS supports different volume types such as **SSD-based and HDD-based** volumes, allowing users to choose performance and cost based on workload requirements. EBS volumes are commonly used for workloads that require **low latency, high performance, and reliable storage**.

Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic